MD SABBIR HOSSEN

Kishoreganj, Dhaka, Bangladesh
+880 1712 459 227 ◇ sabbir.hossen@bu.edu.bd
GitHub ◇ Personal Website ◇ Google Scholar ◇ LinkedIn

RESEARCH INTERESTS

Foundation and Multimodal AI

- Natural Language Processing (NLP)
- Large Language Models (LLMs)
- Computer Vision (CV)
- Vision-Language Models (VLMs)

Efficient and Resource-Constrained AI

- Efficient Deep Learning Architectures
- Low-Resource and Multilingual Language Processing
- Model Compression and Optimization
- AI for Low-Resource Settings

EDUCATION

Bangladesh University — Mohammadpur, Dhaka-1207, Bangladesh
B.Sc in Computer Science and Engineering — *January 2020 - January 2024*
CGPA: 3.66/4.00

Thesis: A Sophisticated Feature Vectorization-Based Stacked Machine Learning Approach for Fake News Detection in Bangla and English.

EXPERIENCES

Research Assistant — *September 2024 - Present*
National Institute of Textile Engineering and Research

- **Cross-Lingual Visual Reasoning in VLMs:** Developing a multilingual visual reasoning benchmark and intervention-based evaluation to assess visual grounding across multilingual and code-switched settings.
- **Low-Resource LLM Adaptation:** Adapting small multilingual LLMs to Bangla while measuring target-language gains and retention of multilingual capabilities across languages and tasks.
- **Small-to-Large LLM Routing:** Developed a routing framework showing that confidence becomes nearly uninformative for reasoning (AUC 0.510) and allocates large-model access unevenly across languages.
- **Quantization of Small LLMs for Bangla:** Benchmarked 0.5B–3B architectures across FP16, INT8, and 4-bit precision; Showed that INT8 retains 97.4% of mean FP16 performance at 3B while reducing memory by 45.0%, whereas 4-bit quantization saves 67.4% with larger task-dependent degradation.

Research Assistant — *May 2023 - July 2024*
Bangladesh University

- **Multilingual Fake-News Detection:** Led the development of a stacked machine-learning framework for Bangla and English fake-news detection, resulting in a first-author publication in a Q1 journal.
- **Dataset Development and NLP Modeling:** Collected and curated datasets and developed Transformer-based models for machine-generated text detection and sentiment analysis, resulting in multiple publications.
- **Research Leadership and Corresponding Authorship:** Conducted literature reviews, prepared manuscripts, and coordinated coauthors, reviewer responses, and revisions across multiple research papers.
- **Research Mentoring:** Mentored three undergraduate research groups on medical imaging and agricultural computer vision, guiding experimental design, model evaluation, and manuscript preparation.

SELECTED PUBLICATIONS

Md. Sabbir Hossen, Fahim Al Farid, Pabon Shaha, Md. Mowahibur Rahman Twake et al. "A Sophisticated Feature Vectorization-Based Stacked Machine Learning Approach for Fake News Detection in Bangla and English." *Social Network Analysis and Mining*, 2025. [**Paper**]

Md. Sabbir Hossen, Md. Saiduzzaman, and Pabon Saha. "Social Media Sentiments Analysis on the July Revolution in Bangladesh: A Hybrid Transformer Based Machine Learning Approach." *In Proceedings of the IEEE 17th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)*, 2025 [**Paper**]

Pabon Saha, **Md. Sabbir Hossen**, Md. Ibrahim Hosen Sojib, and Sanjida Akter et al. "StackTrace-AI: Identifying Generative AI Text Origins using Ensemble Learning." *In Proceedings of the IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, 2025 [**Paper**]

Md. Sabbir Hossen and Md. Saiduzzaman. "TransCNN: A Hybrid CNN-Transformer Synergy for Reliable Deepfake Forensics." *In Proceedings of the IEEE 28th International Conference on Computer and Information Technology (ICCIT)*, 2025 [**Paper**]

Eshat Ahmed Shuvo, Md Shuvon, Nazmul Sarkar, and **Md. Sabbir Hossen** et al. "Optimized Hybrid Cascaded Approach for Accurate Oral Cancer Detection in Histopathology Images Using Deep CNNs." *In Proceedings of the IEEE 2nd International Conference on Next-Gen Computing, IoT and Machine Learning (NCIM)*, 2025 [**Paper**]

Md. Emon Akter Sourov, **Md. Sabbir Hossen**, and Pabon Shaha et al. "Explainable Machine Learning Framework for Cardiovascular Disease Diagnosis and Prognosis." *In Proceedings of the IEEE International Research Conference on Smart Computing and Systems Engineering (SCSE)*, 2026 [**Paper**]

Md. Sabbir Hossen, Md. Saiduzzaman, Pabon Shaha, and Mostofa Kamal Nasir. "Jellyfish Species Identification: A CNN-Based Artificial Neural Network Approach." *In Proceedings of the IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, 2025 [**Paper**]

Md. Sabbir Hossen, Pabon Saha, and Md. Saiduzzaman. "A Hybrid Machine Learning Approach Utilizing CNN Feature Extraction with Traditional Classifiers to Identify Strawberry Leaf Diseases." *In Proceedings of the IEEE 4th International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025 [**Paper**]

Md. Sabbir Hossen, Pabon Shaha, and Md. Saiduzzaman et al. "An Explainable AI Driven Machine Learning Approach for Maternal Health Risk Analysis." *In Proceedings of the IEEE 27th International Conference on Computer and Information Technology (ICCIT)*, 2024 [**Paper**]

SUBMITTED MANUSCRIPTS

Md. Sabbir Hossen, Anichur Rahman, Pabon Shaha, and Airin Afroj Aishi, et al. "Who Pays for Efficient Inference? Confidence-Based Small-to-Large LLM Routing Across Tasks and Languages" *[Submitted to ACL Rolling Review (ARR)]*, 2026 **[Paper][Code →]**

Md. Sabbir Hossen, Anichur Rahman, Pabon Shaha, Andrew H. Sung, and Md Shohel Raha. "How Much Can We Compress Small LLMs? Quantization Trade-offs for Low-Resource Bangla Language Understanding." *[Submitted to ACL Rolling Review (ARR)]*, 2026 **[Paper] [Code →]**

Md. Tanvir Ahmed Shaikat, Pabon Shaha, **Md. Sabbir Hossen**, S M Rifatur Rana et al. "Automated Scoring of Bangla Student Responses: An Efficient and Explainable Deep Learning Approach," 2026 *[Submitted to a peer-reviewed Journal]*

Pabon Saha, **Md. Sabbir Hossen**, Mostofa Kamal Nasir, Anichur Rahman et al. "Catching the Bots: A Transformer-Based Ensemble Learning for Machine-Generated Text Detection," 2026 *[Under Review in a peer-reviewed Journal]*

Md. Sabbir Hossen, Eshat Ahmed Shuvo, Shibbir Ahmed Arif, Pabon Shaha, and Anichur Rahman et al. "An Efficient Deep Learning Framework for Brain Stroke Diagnosis Using Computed Tomography (CT) Images," 2025 *[Under Review in a peer-reviewed Journal]*

Eshat Ahmad Shuvo, Wahidur Rahman, Pabon Shaha, and **Md. Sabbir Hossen** et al. "Optimized Hybrid Approach for Early Detection of Alzheimer's Disease Using Machine Learning and Deep Learning Techniques," 2025 *[Under Review in a peer-reviewed Journal]*

Mohammad Shohel Parves, **Md. Sabbir Hossen**, Pabon Saha, and Bikash Kumar Paul et al. "ConvNet9: A Cutting-Edge Customized Convolutional Neural Network Model to Identify Potato Leaf Disease with Web Application," 2025 *[Under Review in a peer-reviewed Journal]*

Md. Sabbir Hossen, Md. Saiduzzaman, Pabon Shaha, and Bikash Kumar Paul. "Attention-Guided Deep CNN for Robust Image-Based Weather Phenomena Classification," 2026 *[Manuscript under review for publication in an IEEE Conference]*

CONFERENCE PRESENTATIONS

Conference Talk TransCNN: A Hybrid CNN-Transformer Synergy for Reliable Deepfake Forensics. *IEEE ICCIT*, 2025

Conference Talk Social Media Sentiments Analysis on the July Revolution in Bangladesh: A Hybrid Transformer Based Machine Learning Approach, *IEEE ECAI*, 2025

Conference Talk A Hybrid Machine Learning Approach Utilizing CNN Feature Extraction with Traditional Classifiers to Identify Strawberry Leaf Diseases, *IEEE QPAIN*, 2025

Conference Talk Jellyfish Species Identification: A CNN-Based Artificial Neural Network Approach, *IEEE ECCE*, 2025

Conference Talk An Explainable AI Driven Machine Learning Approach for Maternal Health Risk Analysis, *IEEE ICCIT*, 2024 [**Best Technical Presentation**]

ACADEMIC SERVICE

Journal Reviewer: Journal of Plant Diseases and Protection (JPDP), 2026

Conference Reviewer: IEEE USBEREIT, 2026

AWARDS & ACHIEVEMENTS

Best Technical Presentation, 27th International Conference on Computer and Information Technology, 2024

1st Runner Up in Project Showcasing, Robo Carnival, BUET, 2023

PROFESSIONAL SKILLS

Programming Languages: Python, C, C++

Python Libraries: NumPy, Pandas, scikit-learn, Matplotlib

ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers

Software Tools: Git, GitHub, Microsoft Office, LaTeX, Mendeley

EXTRA-CURRICULAR ACTIVITIES

iTech Expo associated with Techfest IIT Bombay, IUBAT *November 2022*
Presented a project on Remote Control Fire Fighter Robot.

Robo Carnival, BUET January 2023
Presented a project on Green Energy E-Bike.

BEC Robo Mania, BUBT January 2023
Presented a project on an Integrated Smart Home.

TEST SCORE

IELTS 7.0